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Kingdom of Saudi Arabia
Ministry of Education
Imam Abdulrahman bin Faisal University
College of Science and Humanities-Jubail
Computer science Department

Software Engineering
Term 1 – 2026/2027

Project Proposal Form

- ✓ Group #:
- ✓ Members of the group:

#	Name	ID	Role
1	Aljazi Alsaiari	2230004865	Leader
2	Roweena Belal	2230005940	Member
3	Noor Alawami	2230000703	Member
4	Forat Alabbas	2230001270	Member

- ✓ **Title of the Project:**

  Food Saver:An Intelligent Food Inventory and Waste Reduction System.

- ✓ **Introduction of the Project - Problem Statement:**

  The issue of food wastage exists across various homes as people sometimes forget about the food items they possess or, otherwise, the food becomes unusable as it reaches its expiration date. For starters, it is not always easy for people to keep track of their supplies and figure out the dishes that they are able to cook with the food they have. FoodSaver offers a new smart technology that helps with food management and food waste reduction. This technology allows users to input food products they own, get information about the expiration dates of their food items, and receive alerts about the food products that are nearing their expiration dates. The application also helps users come up with recipes. Thus, these three aspects of food management technology: food inventory management, expiration alerts, and recipes, allow users to cut down food waste.

  However, even with good tracking, households often end up with surplus food that they cannot consume before it expires, while other individuals and families in the same community are in need of food. Currently, there is no simple way to connect these two groups, so edible food is thrown away. FoodSaver addresses this gap by allowing users to donate surplus food and by matching donations with nearby request based on location, food type, and quantity. To keep users engaged and encourage long term participation, the application also includes a recipe sharing community and a rewards system that recognizes completed donations and volunteer contributions.

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- ✓ **Objectives (Product Characteristics and Requirements):**
  - Create a prototype that is easy for users to use in order to reduce food waste in households.
  - Users should be allowed to keep track of their food inventory and of the expiration dates.
  - Set up notifications and reminders for food items that are approaching their expiration date.
  - Suggest recipes automatically based on the ingredients available in the user's food inventory, prioritizing items that are closest to their expiration date.
  - Allow users to donate any food that they have in excess to those who need it.
  - Include the ability to search and filter according to food category, donation type, location, and quantity.
  - Match up users with nearby food donations and food requests.
  - Create a community in which users can share, discover, and rate recipes.
  - Create a reward system for users who completed volunteer hours and donations.
  - Design an easy to use and user friendly interface.

- ✓ **Project end user:**

The households and individuals who want to reduce food waste and have additional food to give away for people who are looking for available food donations. Moreover, the cooks that would like to share, discover, and review recipes.

- ✓ **Summary of Project Deliverables:**

Food Saver is an intelligent application prototype designed to reduce food waste by connecting people who have surplus food with those who need it promoting a more sustainable and giving lifestyle users can customize the app based on their personal preferences, location, and the type of food they wish to donate or request it provides a built in expiry tracking system that notifies users when their stored food items are nearing expiration, helping them take timely action before the food go to waste. Also,the app offers search and filtering by food category, donation type, pickup location and available quantity making it easy for users to find or offer suitable donations nearby.

Food Saver recommends relevant donation and request opportunities based on user location, food type, and real time availability,supported by simulated local data listings that ensure smooth interaction flow. Additionally, the prototype features an active community component where users can share, discover, and rate recipes, with reviews by experienced home cooks to ensure quality**.** The prototype helps users donate surplus food, minimize waste, and actively support their community it further supports smart notifications, reminders, and a rewards system that recognizes user contributions by counting completed donations toward official volunteer hours with a clean and user friendly interface, Food Saver is designed for individuals committed to reducing food waste, enabling them to save food, help others, and create a meaningful positive impact with ease.

Project Deliverables include:

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- A high fidelity interactive mobile application prototype (iOS/Android UI/UX screens and core navigation).
- Community and Recipe Sharing module**:** A dedicated space for users to share recipes and view steps reviewed by experienced home cooks.
- Core interactive modules: Food stock tracking, expiration alerts, category-based search and filtering, and donation matching interface.
- Smart recipe suggestion module: A screen that recommends recipes based on the ingredients currently stored in the user's inventory, giving priority to items nearing their expiration date.
- Community and Rewards module with volunteer hours calculation screen for completed donations.
- Complete project documentation: Software Project Management Plan (SPMP), Software Requirements Specification (SRS), System Design Specification (SDS) and Software Test Plan (STP).

✓ **Project Success Criteria:**

The project will be considered successful based on the following:

- Ease of use and smooth navigation through the application.
- Stable performance with minimal errors or crashes.
- Timely expiration notifications that help users use their food before it expires.
- The ability of users to create food donations and find suitable nearby requests easily through search and filtering.
- User satisfaction with the recipe suggestions and the recipe community.
- Reduction in the time users spend tracking their food and deciding what to cook.

✓ **Solution Methodology:**

The project will follow an Agile development methodology using Scrum. The system will be developed incrementally through a series of short sprints. The main features will be divided into smaller tasks and prioritized in a product backlog. Each sprint will focus on developing a specific set of features, such as food inventory and expiration tracking, notifications and reminders, food donation and search, and the recipe community. At the end of each sprint, the developed features will be tested and reviewed to identify improvements and changes needed for the next sprint. This approach allows FoodSaver to be continuously improved based on feedback and changing requirements.

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✓ **Signatures of group members**

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1	Aljazi alsaiari	2230004865	
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4	Forat Alabbas	2230001270	

Date: 20/09/2026

---------- **For instructor use only** ----------

Approved Not approved Date: …….

Comments:

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